Effects of Working Memory on Morphological Priming

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Abstract

Research in morphological word processing attempts to study the relationship between morphological decomposition and semantic transparency by comparing (i) semantically *transparent* (morphologically and semantically related: misty-MIST), (ii) semantically *opaque* (morphologically related but semantically unrelated: mister-MIST), and (iii) *form-related* (morphologically and semantically unrelated: mistake-MIST) word pairs under a masked priming paradigm. Owing to mixed results found in the literature, theoretical models differ on whether the basis of morphological decomposition is morphological (morpho-orthographic account), or semantic (morpho-semantic account). It has been assumed that masked priming typically used in these studies suspends conscious, attentional resources and hence reflects automatic priming effects; however, recent research shows that there may be effects of latent attentional resources in this paradigm. To resolve the discrepancies in the literature, the present study examined the automatic effects of morphological priming by embedding a secondary memory load task into the traditional masked priming paradigm. Data from 40 university students showed significant priming effects for transparent words under both high and low memory loads, but no priming for opaque and form-related words, implying that masked priming does not effectively control for conscious processing, and that the effects of semantic transparency are automatic in morphological decomposition, thus providing strong evidence for the morpho-semantic account.

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Introduction

There are certain effects observed in the speed with which people recognise words. The study of isolated visual word recognition processes mainly focuses on creating a psychological model that can account for these observed effects. These models theorize the existence of a mental dictionary or lexicon that holds representations of words and their associated features (Coltheart, 2006; Seidenberg, 2005). This lexicon is accessed every time a person reads and can influence the process of visual word recognition. Psycholinguistic researchers often study the morphology of words in particular to attain insight into the structure of the mental lexicon. The field of morphological research spans decades of study and has evolved many specific, technical terms which are important to clarify at the outset of this paper. In this section, I will first present a brief overview of these terms, followed by a summary of the empirical data in morphological studies as well as a review of the current models that attempt to account for these effects.

Morphology is the study of internal structure of words and the relations between them. A *morpheme* is the smallest unit of language: it is a word that has meaning and cannot be further broken down into any meaningful constituents, such as ‘mist’. A morpheme can be deployed as a stem to create other, *morphologically related words* by the addition of other morphemes, such as affixes (‘be’ to create ‘bemist’) or suffixes (‘y’ to create ‘misty’). The *lexical decision task*, or LDT, measures how quickly a person decides whether a given set of letters together make up a real word (‘MIST’) or a nonsense word (‘TIST’). The basic assumption of this task is that faster reaction times (RTs) reflect a more efficient processing of the word in the mental lexicon. Much research utilizing the LDT has shown that the presentation of morphologically related words prior to a target stem word can facilitate a faster reaction to the stem as compared to the presentation of a different, unrelated morphemic construction. In other words, the presentation of a word that is morphologically

related to the target (e.g. ‘misty’) prior to the target itself (e.g. ‘MIST’) increases the speed of recognition of the target word compared to the prior presentation of an unrelated morphologically complex word (e.g. ‘lanky’). The prior presentation of the morphologically related word is assumed to facilitate or *prime* the recognition of the target – hence, this phenomenon is known as the *morphological priming effect*.

The morphological priming effect is presumed to occur due to the decomposition of the morphologically related word (‘misty’) into its stem (‘mist’) and suffix (‘y’). One of the earliest models of word morphology is Taft and Forster’s (1975) *morphological decomposition model*, as seen in Figure 1, which suggests that people process morphologically complex words by dividing them into their constituent morphemes and checking the lexicon for the stem word. According to this model, morphological priming occurs because the decomposition of the prime word ‘misty’ yields the lexical entry for the stem morpheme ‘mist’, hence facilitating the later recognition of ‘MIST’. This model stands in contrast to the *whole word processing model* wherein morphologically complex words are recognized as they are, without any decomposition (Butterworth, 1983 as in Frisson et al., 2008). However, a majority of research has converged around the morphological decomposition model, given two reasons: first, it is counterintuitive to our use of language to have different lexical entries for related words such that “the semantics are duplicated for each separate variant” (Marslen-Wilson et al., 1994, p. 31) and second, it is difficult to reconcile the whole word processing model when confronted with the vast empirical evidence supporting morphological priming effects (Amenta & Crepaldi, 2012).

However, it is also evident that the priming effect of ‘misty’ to ‘MIST’ is more complex than Taft and Forster’s (1975) simple morphological decomposition, owing to the several different types of properties that provide relatedness between the two words: for instance, they both share a similar composition of letters (*orthographically* related), sounds

(*phonologically* related), and meaning (*semantically* related). To this end, much research in the field of morphology applies itself to better understand morphological relatedness by detaching these properties from each other (Drews, 1996). This, as I will discuss in the next section, has been primarily achieved by using primes that are related to the target only on certain properties and not others, then comparing RTs among them.

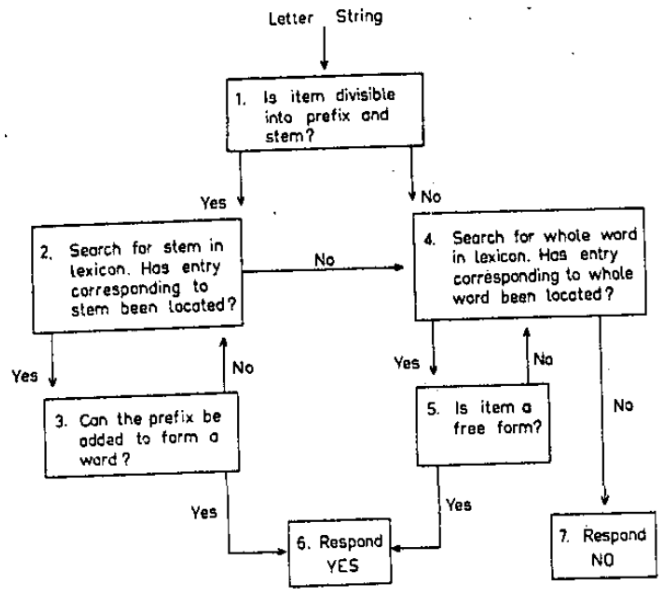


Figure 1. The morpho-orthographic account of morphological decomposition as in Taft and Forster (1975).

Processes in Morphological Priming

The differential effects of semantics and orthography in word morphology are studied by comparing different types of prime-target relationships. These are: (i) semantically *transparent* word pairs like 'misty-MIST' which are semantically, morphologically and orthographically related, (ii) semantically *opaque* word pairs like 'mister-MIST', which are morphologically and orthographically related but semantically unrelated, and (iii) *form-related* word pairs like 'mistake-MIST', which are orthographically related, but morphologically and semantically unrelated. These effects are measured against a baseline condition of morphologically, orthographically and semantically *unrelated* word pairs, like 'lanky-MIST'. For the purposes of clarity in this paper, I have used examples with the same

stem (MIST); however, in the course of an experiment a target is presented only once in each condition, as shown in Table 1, to prevent any additional facilitation effects.

In order to uncover semantic effects in the morphological decomposition process, transparent and opaque words are compared since they differ from each other only semantically. That is, both ‘misty’ and ‘mister’ can be decomposed to the stem word ‘MIST’, but only ‘misty’ shares any semantic relatedness with it – therefore, any difference in the reaction time (RT) to a transparent prime (here, ‘misty’) as compared to an opaque prime (‘mister’) can be attributed to the property of semantic relatedness. In order to uncover morphological effects, we compare opaque and form-related word pairs, since they are only differentiated by morphology. Neither ‘mister’ nor ‘mistake’ are semantically related to ‘MIST’; however, unlike ‘mistake’, ‘mister’ can be decomposed (‘mist’ + ‘er’)[1] – therefore, any difference in the reaction time (RT) to an opaque prime (‘mister’) as compared to a form-related prime (‘mistake’) can be attributed to morphological decomposition.

Table 1

Example stimuli for test conditions.

Condition	Example		Orthographic Relation?	Morphological relation?	Semantic Relation?
	Prime	Target			
Transparent	walked	WALK	Yes	Yes	Yes
Opaque	listen	LIST	Yes	Yes	No
Form-Related	freeze	FREE	Yes	No	No
Unrelated	marshy	STUN	No	No	No

There has been found a consistent priming effect for transparent word pairs (misty-MIST), and little to no priming effect seen for form-related pairs (mistake-MIST) across the

[1] We do not decompose ‘mistake’ into ‘mist’ and ‘ake’ because ‘ake’ cannot be added to another stem to form a word and hence is not a legal suffix, unlike ‘er’.

visual word recognition literature (Andrews & Lo, 2013; Marslen-Wilson et al., 2008; Morris et al., 2007; Rastle et al., 2004). As such, we can firmly conclude that morphological priming does not benefit from mere orthographic overlap. The question, then, arises around the degree of benefit provided by morphology to priming in the semantic transparency condition. The answer to this lies in the opaque priming condition (mister-MIST) which provides morphological but not semantic relatedness. However, the empirical data regarding opaque priming has yielded mixed results: some studies report no effect of opaque priming, while others have shown a compelling significant effect of the same. This conflicting data has given rise to differing accounts of the processes involved in morphological priming. I will present a summary of these accounts below, synthesized into two broad categories of morphological priming models: the morpho-orthographic and morpho-semantic views, each of which differ on whether decomposition is conducted on the basis of morphology or semantics.

Morpho-orthographic account

Many studies have found significant facilitation for opaque pairs – that is, 'MIST' is recognized just as fast when preceded by 'mister' as it is when preceded by 'misty' (Feldman et al., 2004; Lavric et al., 2007; Rastle et al., 2000, 2004; Whiting et al., 2017). This *morpho-orthographic* account posits that the decomposition of morphemes does not access the semantic information of a word, but restricts itself to the orthographic form of the constituent morphemes, particularly when the prime is hidden under masked priming (Lavric et al., 2007) and short stimulus-onset-asynchrony (SOA) conditions (Feldman et al., 2004; Rastle et al., 2000). Though this effect disappears when the primes are visible to subjects, the observation of opaque priming effects under masked conditions make a strong case for an early, morphological decomposition process.

According to this view, 'mister' and 'misty' both facilitate the recognition of 'MIST' under masked conditions because they both retrieve the stem 'mist' upon decomposition.

This decomposition occurs early in the word recognition process, and operates on "any printed word that contains a stem and an affix" (Rastle et al., 2004, p. 1093) – hence, it is entirely unaffected by semantic characteristics. These priming effects are known to be purely morphological in nature owing to the fact that form-related primes do not produce the same advantage in recognizing the target word, despite having orthographic overlap with it. Similar to the Taft and Forster (1975) model in Figure 1, this account supports an immediate decomposition of a given word into its constituent morphemes, rather than compute it as a whole word before decomposition. On this view, then, opaque primes are equally as effective as transparent primes since both benefit from morphological decomposition, unlike a form-related word (Davis & Rastle, 2010).

Morpho-semantic account

Other studies have demonstrated a facilitatory effect of semantic transparency and no effect from opaque priming (Diependaele et al., 2009; Feldman et al., 2009; Marslen-Wilson et al., 1994; Morris et al., 2007; Rastle et al., 2000) – that is, 'MIST' is recognized fastest when preceded by 'misty' as compared to 'mister'. These results are explained by what is known as the *morpho-semantic account*, which posits that this facilitation results from the congruency between the prime and its activation of the semantic meaning of the prime's stem. Indeed, many studies using a variety of paradigms, like delayed repetition priming (Marslen-Wilson & Zhou, 1999), cross-modal priming (Marslen-Wilson et al., 1994), long SOA intervals (Rastle et al., 2000) and even masked priming (Diependaele et al., 2005, 2009; Feldman et al., 2009; Giraudo & Grainger, 2001; Morris et al., 2007) have found this effect.

One of the earliest models of the morpho-semantic priming effect (Marslen-Wilson et al., 1994) put forth the theory that while transparent words are morphologically decomposed, opaque words are recognized as whole words because their lexical entries are separate from their apparent stem. As seen in Figure 2, the opaque word 'department' is a stem for

'departmental' as they are semantically related, but 'department' and 'depart' are separate lexical entries. Thus, morphological decomposition is conducted on the basis of semantics on this view.

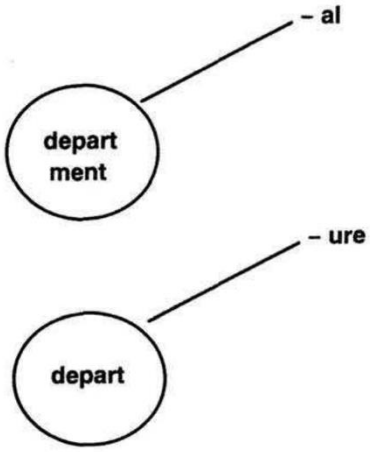


Figure 6. Lexical entries for a cognitively monomorphemic word (*department*), shown here in combination with the affix *-al*, and for a morphologically complex form (*departure*).

Figure 2. The morpho-semantic model of lexical entries of words as in Marslen-Wilson et al., 1994.

Building on this model, Giraudo and Grainger (2001) emphasized the role of whole-word processing in this morpho-semantic account – they put forth the idea that a prime is first seen as a whole word, and only further decomposed into its constituent morphemes based on semantic transparency. Thus, using the examples in Figure 2, Giraudo and Grainger (2001) would argue that the system checked for a corresponding stem for 'departure', which led to 'depart', and these representations activated each other in the lexicon. However, since 'department' has no stem, it is seen as a whole word. The basic assumption is that we leverage on a "systemic correspondence of form and meaning" (Diependaele et al., 2012, p. 315) between the prime and the stem for lexical access rather than the morphological construction, resulting in a processing advantage. Therefore, transparent word pairs (misty-MIST), rather than the opaque pair (mister-MIST) allow the target word to be recognized faster due to semantic overlap.

It is difficult to reconcile the discrepancies between the morpho-semantic and the morpho-orthographic accounts. The conflict arises when we consider what lexical system to adhere to when trying to explain both these effects within the same model: researchers are grappling with the question of whether morphological decomposition is conducted on the basis of solely orthography or semantics. These conflicts have been explained away as a consequence of methodological differences by the proponents of the morpho-orthographic account (Davis & Rastle, 2010) and as a result of inconsistent stimulus properties by the proponents of the morpho-semantic account (Feldman et al., 2009). Further studies have made cases for task demands (Duñabeitia et al., 2011; Paterson et al., 2011), demographic differences in vocabulary (Andrews & Lo, 2013), and trends derived as a linear function of morphological structure and semantic transparency (Morris et al., 2007; see Amenta & Crepaldi, 2012 for further review). It is the latter consideration that has given rise to a third account of these mixed morphological priming effects, which attempt to accommodate both morpho-orthographic and morpho-semantic results.

Hybrid Accounts

To summarize, *morpho-orthographic* models posit a purely morphological basis of decomposition, which occurs prior to a whole-word recognition of the prime; as such, there should be an equivalent priming effect of transparent and opaque word pairs (see Figure 3A). *Morpho-semantic* models posit a process of decomposition that first assesses semantic transparency of the prime as a whole word, which implies that there should be no priming for opaque primes; that is, there is an equivalent lack of priming for opaque and form-related word pairs (see Figure 3B). However, there may be a case where opaque priming effects are not entirely equivalent to either transparent or form-related pairs – instead, they fall somewhere in the middle (see Figure 3C).

These hybrid models remove the assumption that semantic and orthographic decomposition routes are independent of each other – instead, they see the morphological priming effect as a sum of semantic and morphological facilitation (Morris et al., 2007). Thus, transparent word pairs are most facilitated due to both semantic and morphological relatedness, opaque word pairs are facilitated to a lesser degree due to a semantic dissimilarity but benefit from morphological relatedness, and form-related pairs have no facilitation due to a complete lack of semantic and morphological relatedness (Diependaele et al., 2009). However, it must be noted that further studies of these models rely on neuroscientific observations and network analyses rather than the traditional behavioural data I have discussed in this paper. Moreover, this research suggests that it is likely that the interdependence of morphology and semantics may be a nonlinear effect (see Feldman & Milin, 2019), and these effects seem to be highly sensitive to individual factors regarding linguistic proficiency (Milin et al., 2017).

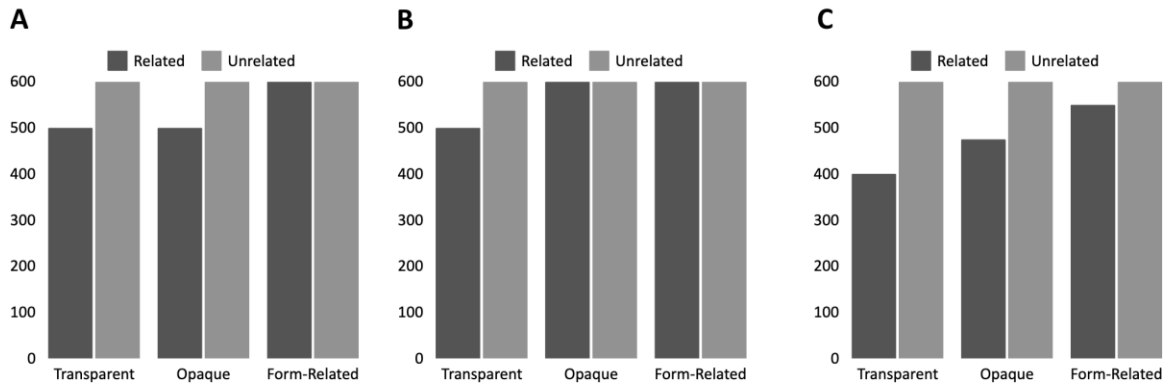


Figure 3. The predictions of each morphological priming model. The morpho-orthographic model (A) predicts equivalent priming between transparent and opaque pairs, the morpho-semantic model (B) predicts an equivalent lack of priming between opaque and form-related pairs, and the hybrid account (C) predicts priming effects for opaque pairs which is stronger than form-related pairs but weaker than transparent pairs.

Question of Automaticity

As we have seen so far, the central point of contention between the morpho-orthographic and the morpho-semantic accounts revolves around the role of semantics in the

early, unconscious stages of visual processing. Subsequently, research in this field focuses on studying unconscious, *automatic* processes rather than strategic, attentional, conscious ones in morphological priming. The debate hence takes on a particularly controversial note given that many of the studies in this field use the masked priming paradigm, which is specifically used to inhibit conscious processes and attentional resources.

Masked Priming Paradigm

In a standard masked priming paradigm, as seen in Figure 4, the participants first see a series of symbols (######) that serve as a forward mask for the prime. The prime itself is briefly presented for durations ranging from 36 to 52ms. The target is then presented for lexical decision, which also serves as a backward mask for the prime. The extremely short duration of the prime, combined with the double masking provided by other visual events, is theorised to prevent conscious processing of the prime; thus, any priming effects seen under masked conditions can be attributed to automatic and unconscious processing (Balota, 1983; K. I. Forster & Davis, 1984). The myriad of studies (Andrews & Lo, 2013; Duñabeitia et al., 2011; Morris et al., 2007; Rastle et al., 2000, 2004) that show effects using the masked priming paradigm serve as substantial evidence that there *is* some sort of automatic, subliminal mechanism in word recognition. However, in the field of morphological priming, it is yet unclear *what* these automatic processes are leveraging on – semantics or orthography? The question yet to be answered is whether, and in what domain, morphological processing is automatic. The discrepancies in the morpho-orthographic and morpho-semantic account exist despite both sides' claims to automaticity: it is prudent, then, to examine more closely the role of automaticity in masked priming.

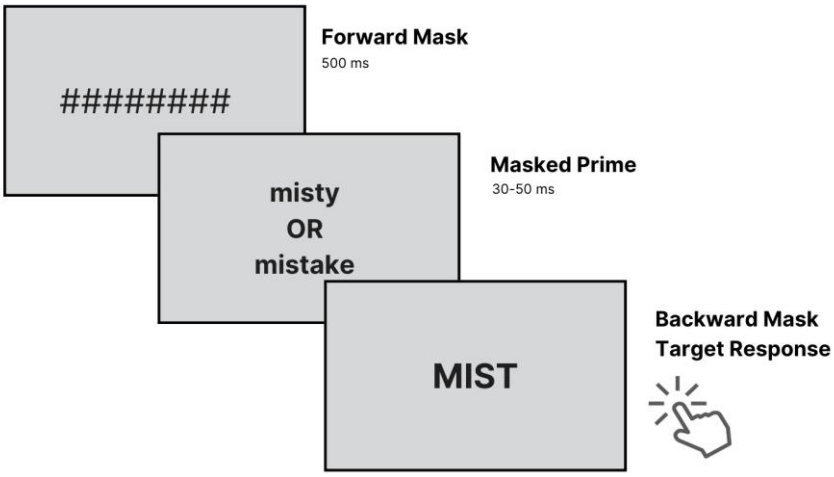


Figure 4. A typical masked priming paradigm.

Role of Working Memory

Questions regarding automatic processing in word recognition have been addressed in the wider net of psycholinguistics by several researchers; in particular, the role of working memory has been emphasized in reading and word recognition processes. The term *working memory* generally refers to a cognitive system which hold "mental representations temporarily available for use" in most conscious tasks (Oberauer et al., 2018, p. 886). Components of working memory have been associated with greater reading speed and comprehension (Dark & Benbow, 1991; Katz et al., 2012; Kolers et al., 1979). Further research has shown that not only does overloading the working memory through attention-dividing tasks affect performance on the LDT by resulting in slower reaction times (Heyman et al., 2015), but also makes it more difficult to perform presumably-automatic processes such as resolving lexical ambiguity (Chein et al., 2011).

The concern, then, arises that the masked priming paradigm might still leave some conscious attentional resources available and leverage on them. Bodner and Stalinski (2008) were the first to implement a between-subjects load task within a masked priming paradigm – they found a significant effect of load and priming, but no interaction. However, they did not control the experiment with a corresponding low load for the participants, which may have

reduced the power of the experiment. Addressing this facet of the study, Perea et al. (2018) tested how the automaticity of masked identity priming would be affected by overloading the working memory: they imposed a secondary working memory task into the traditional masked priming paradigm using a within-subjects design. Perea et al. (2018) used primes that were either identical (duck-DUCK) or unrelated (wall-DUCK) to the target. In Experiment 2, they conducted this dual task procedure by presenting a string of numbers, which was either simple (low load, e.g. 55555) or complex (high load, e.g. 49857), for the participants to remember and later, make a recognition response. On each trial, as seen in Figure 5, participants were first shown the string of numbers (55555 or 49857), followed by a forward mask (#####), a masked prime for 50ms (duck or wall), a target to respond to (DUCK), and finally make a recognition response (5?) to a single digit which may or may not have been present in the string of numbers that was shown earlier.

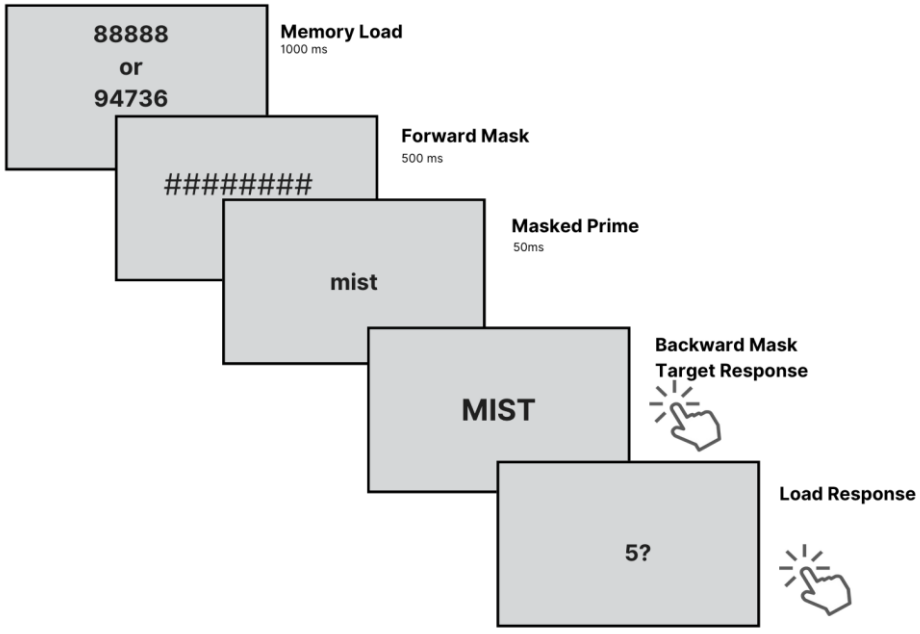


Figure 5. The masked priming paradigm embedded with a working memory task as in Perea et al. (2018).

The results, as seen in Figure 6, showed that there was a significant main effect of load, which means that participants were indeed slower in the high load compared to the low

load condition. There was a significant main effect of relatedness, meaning that priming occurred across load conditions. However, there was no interaction effect, which means that identity priming is not affected by load. This study conclusively shows that identity priming is an automatic phenomenon. Importantly, Perea's et al. (2018) study – particularly the main effect of load – demonstrates that the overloading of working memory is methodologically different from only a masked priming paradigm in that it actively inhibits certain mental processes and ensures a more holistic automaticity in response times.

Table 2 Mean response times (in ms) and percentage error rate (in parentheses) across conditions for words and pseudowords in Experiment 2

	Low memory load	High memory load
Word trials		
Identity	690 (2.4)	712 (4.0)
Control	725 (3.6)	752 (2.8)
Priming	35 (1.2)	40 (−1.2)
Nonword trials		
Identity	807 (4.7)	847 (5.3)
Control	827 (5.8)	843 (5.7)
Priming	20 (1.1)	−4 (0.2)

Note. *Priming* refers to the difference between the control and identity conditions

Figure 6. Results from Experiment 2 in Perea et al. (2018).

Present Study

The present study aims to apply the Perea et al. (2018) paradigm to the study of morphological priming. More specifically, this study attempts to resolve the discrepancy between the morpho-orthographic versus morpho-semantic accounts with regard to word recognition in transparent, opaque and form-related pairs of words. I aim to achieve this by actively overloading the working memory using a recognition task and thereby provide further insight into the automaticity of morphological decomposition. By using a 2 x 2 factorial analysis, which manipulates cognitive Load (High, Low) and prime-target Relatedness (Related, Unrelated) across three different prime types (transparent, opaque,

form-related), I aim to examine the effects of automaticity on the priming generated by each of the prime types.

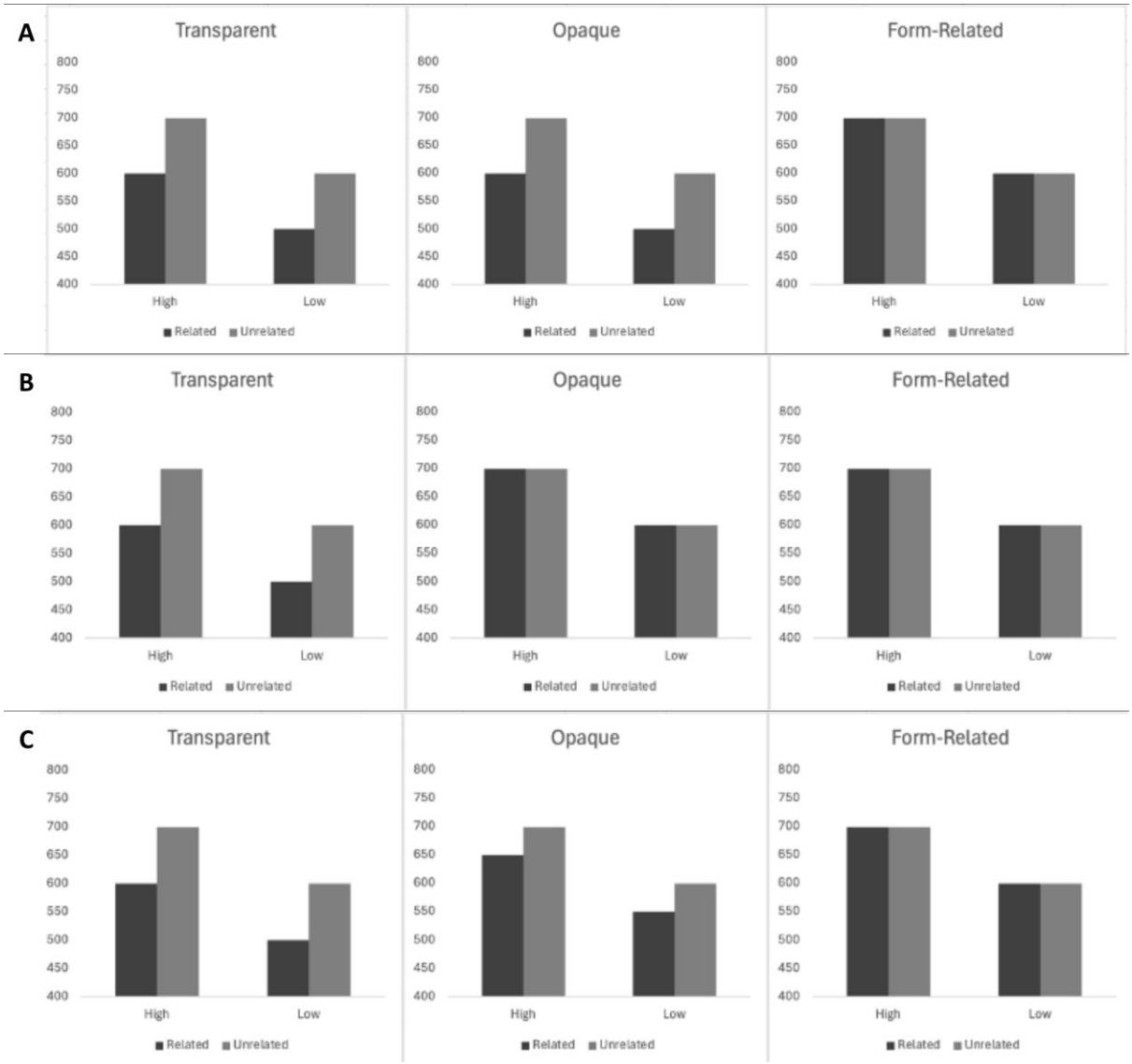


Figure 7. The predicted results of the present study by each morphological priming model. In all models, the RT is slowed for the high load. The morpho-orthographic model (A) predicts equivalent priming between transparent and opaque pairs across load, the morpho-semantic model (B) predicts an equivalent lack of priming between opaque and form-related pairs across load, and the hybrid account (C) predicts priming effects for opaque pairs which is stronger than form-related pairs but weaker than transparent pairs across load.

The predictions of each of the models should follow the same pattern from in Figure 3 above, under both high and low load conditions. The predictions of each model under a load condition is presented in Figure 7. In particular, the opaque condition, around which most of the debate in the current literature revolves, should be examined – if an opaque priming effect

as seen in Figure 7A is observed under a high load condition, we can conclude that the morpho-orthographic account stands, and morphological priming is automatic; however, a lack of opaque priming effect under the high load condition as seen in Figure 7B will support the morpho-semantic view that morphological decomposition benefits from semantic properties under automatic conditions. An opaque priming effect between the transparent and form-related conditions, as seen in Figure 7C, will provide support for the hybrid account.

Methods

Participants

A total of 40 participants (23 female, 17 male; mean age = 22.05 years, *SD* = 2.23) took part in the study. All were native English speakers with normal or corrected-to-normal vision. Participants provided written informed consent and were compensated $12.5 for their time. Each session lasted approximately 65 minutes.

Materials

The stimuli were drawn from materials reported in Rastle et al. (2004), Marslen-Wilson et al. (2008), Morris et al. (2007), Beyersmann et al. (2012), Zhang et al. (2017) and Whiting et al. (2017). Within this list, duplicate words as well as primes or targets which did not have an entry in the English Lexicon Project (ELP) database (Balota et al., 2007) were excluded.

Using Match (van Casteren & Davis, 2007), 300 prime-target pairs were chosen. One-third (100) of these pairs were morphologically related and had a semantically and orthographically transparent relationship (referred to as the transparent condition; misty-MIST), one-third were not morphologically related and had a semantically opaque relationship (opaque; mister-MIST), and one-third were orthographically but not semantically or morphologically related (form-related; mistake-MIST). Following Morris et al. (2007), the prime target pairs were matched across the three conditions on length of the prime, $F(2,297) = .774$, $p = .462$, frequency of the prime, $F(2,297) = .446$, $p = .641$, length of the prime suffix (or in the case of the orthographically related words, the characters at the end of the word that did not overlap with the target), $F(2,297) = .151$, $p = .860$, frequency of the target, $F(2,297) = .1971$, $p = .141$, length of the target, $F(2,297) = 2.373$, $p = .095$, and neighborhood size of the target, $F(2,297) = .051$, $p = .951$. Following Rastle et al. (2004) and Beyersmann (2012), the prime-target pairs were also matched on the number of syllables of the target, $F(2,297)= .00$,

p =1.0, and the latent semantic analysis (LSA) values of the pair of words, $F(2,297)$ = 226.798, $p < 0.001$ (see Table 1 for average values on all properties). The significant value of LSA reflects the significant differences in semantic relatedness of the word pairs across three conditions.

Table 2

Stimulus properties matched across test conditions.

	Transparent		Opaque		Form-Related	
	Mean	SD	Mean	SD	Mean	SD
Prime Length	6.54	1.0	6.54	1.1	6.38	1.1
Prime Frequency	2.06	0.6	2.14	0.7	2.14	0.7
Suffix Length	2.20	0.8	2.23	0.8	2.26	0.8
Target Length	4.32	0.7	4.37	0.8	4.14	0.8
Target Frequency	2.71	0.7	2.53	0.6	2.58	0.7
Target Neighbourhood	6.62	4.9	6.50	4.9	6.72	4.9
Target Syllables	1.14	0.3	1.14	0.3	1.14	0.3
LSA	0.45	0.2	0.12	0.1	0.11	0.1

Stimuli lists underwent counterbalancing. In each condition, the 100 items were randomly divided into four groups of 25 prime-target pairs. There was no significant difference on the previously mentioned characteristics between each subset ($p > 320$). Next, the primes were randomly shuffled within each subset of 25 items in order to create unrelated prime-target pairs. Targets that were preceded by a related word under a high working memory load in one list were preceded by a related word under a low load, an unrelated word under a high load, or unrelated word under a low load in the others. Each subject saw only each target once; therefore no subject saw any given target in both a high and low load, or preceded by both a related and unrelated item.

Using Wuggy (Keuleers & Brysbaert, 2010), 300 nonword targets were also created. Each nonword target was preceded by a morphologically complex real word prime, derived by changing one letter of the nonword stem to form a word, before attaching a suffix to form

a morphologically complex word prime (crispy-CRISK). Similar to the stimuli word lists, nonword lists underwent counterbalancing. One fourth (75) of the nonword targets were preceded by the primes from which they had been derived under high load, one fourth were preceded by the primes from which they had been derived under low load, one-fourth were preceded by unrelated primes under high load and finally, one fourth were preceded by unrelated primes under low load.

Procedure

Participants were tested individually in sound-attenuated cubicle. Stimulus presentation and data recording was controlled for by the DMDX software (K. Forster & Forster, 2003) running on PC computers. Subjects first provided demographic information, including their age, gender, and year of education. Participants were informed that they would be presented with a series of letter strings and required to decide as quickly and accurately as possible whether or not each string was a word. Subjects were told to press the right shift key for words and 'yes' responses to the load recognition task and the left shift key for nonwords and 'no' responses to the load recognition task.

Participants were given 20 practice trials (2 Transparent, 2 Opaque, 2 Form-Related, 4 Unrelated, and 10 nonwords), followed by 600 experimental trials that were randomly presented. Breaks were given every 60 trials. Each trial consisted of a memory task embedded within the masked priming paradigm, similar to Perea et al. (2018) as seen in Figure 3 above. However, we replaced the set of five numbers with four symbols symbols in order to ensure a full working memory load; the low memory load was a string of four similar symbols (÷÷÷÷) and the high memory load was a string of four different symbols (÷$@!). The symbols '#', '+' and '?' were not used in the load task to prevent confusion with the forward mask, fixation point and target response screen. The target response screen showed the target flanked by two question marks (? ÷ ?) to prevent similar confusion.

On each trial, participants first saw a fixation point (+) for 400ms, followed by the symbol string (%%%% or @!%=) for 1000ms. Next, the masked priming paradigm was conducted, with a 500ms forward mask comprising of 13 hash marks (#############), immediately followed by the prime in lowercase letters for 42ms, and the target in uppercase until a response was made or for 3000ms. Lastly, participants were asked to make a decision about whether the target on the screen (? % ?) was present in the original symbol string. The sequence of each trial has been summarized in Figure 8.

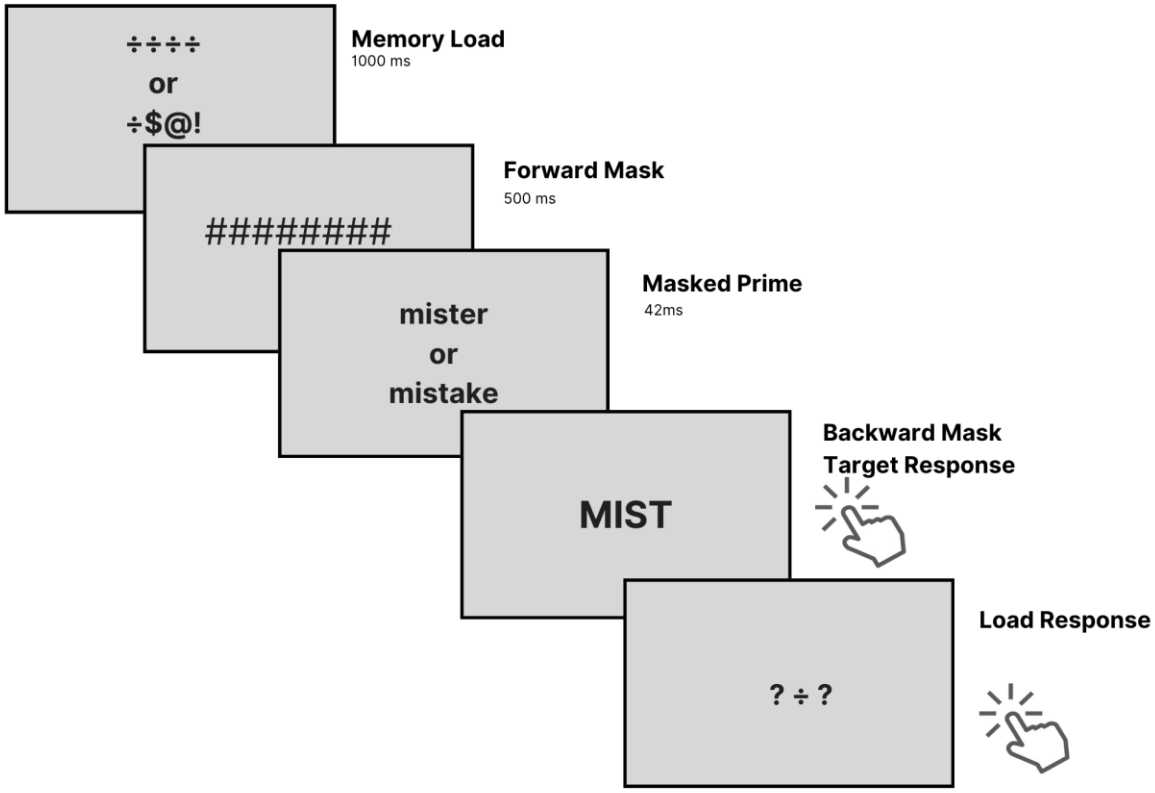


Figure 8. Sequence of visual events in each trial of the present study.

Results

Reaction times (RTs) were obtained and cleaned in several steps. Firstly, error responses were removed and response times that did not fall within the range of 200 to 3000ms were removed. This constituted 4.77% of the data. Next, standardized RT scores for each participant were derived, and those which were greater than or lesser than 2.5 SDs away from each subject's mean were considered outliers and removed. This constituted an additional 2.95% of RT data, resulting in a total of 7.72% data removed after cleaning. A repeated measures ANOVA by participants and items were subsequently implemented.

Table 3

Mean RT and accuracy by condition

	Low			High		
	Mean RT	SD	Accuracy	Mean RT	SD	Accuracy
Transparent						
Related	648.88	77.0	0.95	719.43	111.6	0.76
Unrelated	676.10	94.7	0.94	747.14	111.4	0.80
Priming	27.22*			27.71*		
Opaque						
Related	689.78	689.8	0.93	746.70	116.6	0.77
Unrelated	681.48	681.5	0.93	751.24	105.0	0.73
Priming	-8.29			4.54		
Form-Related						
Related	695.31	87.9	0.94	734.94	101.2	0.77
Unrelated	698.27	70.3	0.90	753.02	102.3	0.76
Priming	2.96			18.08		
Nonword						
Related	753.06	68.6	0.93	809.72	100.0	0.71
Unrelated	765.76	82.7	0.92	806.37	103.1	0.74
Priming	12.70			-3.35		

Analyses

A 2 (Load: High, Low) x 2 (Relatedness: Related, Unrelated) within-subjects ANOVA was conducted for each of the prime types (Transparent, Opaque and Form-Related). These analyses were conducted separately for participant-wise (F_p) and item-wise (F_i) data. Reaction time and accuracy data are presented in Table 3. The priming effects have been graphically represented in Figure 10 below.

Transparent

The ANOVA analysis, as seen in Figure 9A, revealed a significant main effect of Load [$F_p(1,39) = 23.37$, $p < .001$, $\eta_p^2 = .29$; $F_i(1,99) = 116.78$, $p < .001$, $\eta_p^2 = .23$] and Relatedness [$F_p(1,39) = 15.11$, $p < .001$, $\eta_p^2 = .04$; $F_i(1,99) = 8.85$, $p < .005$, $\eta_p^2 = .03$]. However, there was no interaction between the two factors [$F_p(1,39) = .01$, $p = .967$, $\eta_p^2 = .00$; $F_i(1,99) = 0.163$, $p = .687$, $\eta_p^2 = .00$].

Opaque

The ANOVA analysis, as seen in Figure 9B, revealed a significant main effect of Load ($F_p(1,39) = 18.58$, $p < .001$, $\eta^2 = .26$; $F_i(1,99) = 63.36$, $p < .001$, $\eta_p^2 = .19$]; however, there was no significant effect found for Relatedness [$F_p(1,39) = .100$, $p = .75$, $\eta^2 = .00$); $F_i(1,99) = 0.01$, $p = .968$, $\eta_p^2 = .00$. There was no interaction between the two factors [$F_p(1,39) = 1.17$, $p = .287$, $\eta_p^2 = .01$; $F_i(1,99) = 0.68$, $p = .412$, $\eta_p^2 = .00$].

Form-Related

The ANOVA analysis, as seen in Figure 9C, revealed a significant main effect of Load ($F_p(1,39) = 11.69$, $p < .001$, $\eta^2 = .16$; $F_i(1,99) = 58.83$, $p < .001$, $\eta_p^2 = .15$];however, there was no significant effect found for Relatedness [$F_p(1,39) = 1.70$, $p = .200$, $\eta^2 = .008$; $F_i(1,99) = 2.21$, $p = .141$, $\eta_p^2 = .00$]. There was no interaction between the two factors [$F_p(1,39) = 1.17$, $p = .197$, $\eta_p^2 = .00$; $F_i(1,99) = 1.80$, $p = .183$, $\eta_p^2 = .00$].

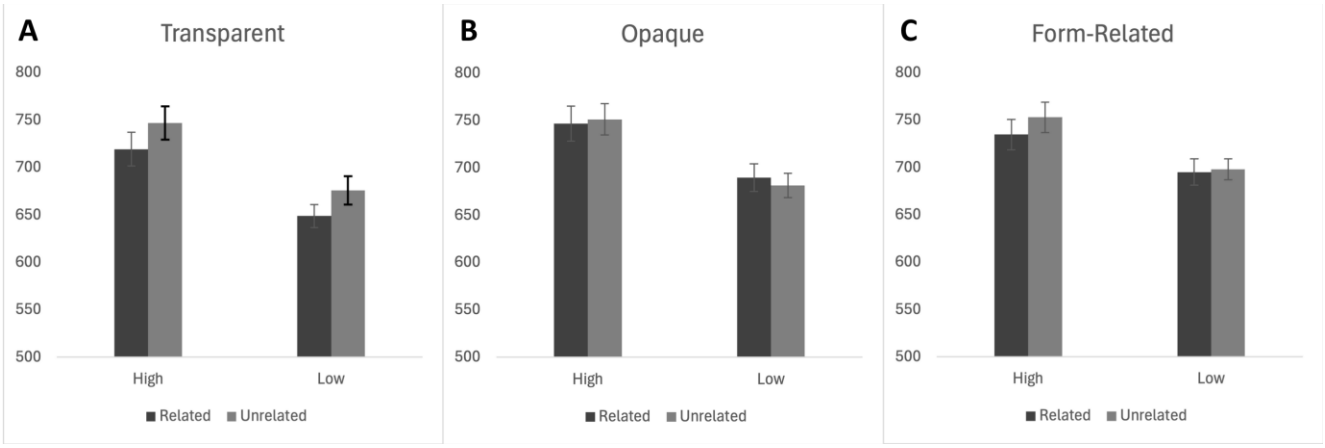


Figure 9. The mean RTs as a function of Relatedness and Prime Type. There was significant effect of priming in both high and low load in the Transparent (A) condition, but no priming in either load condition for the opaque (B) or form-related (C) conditions.

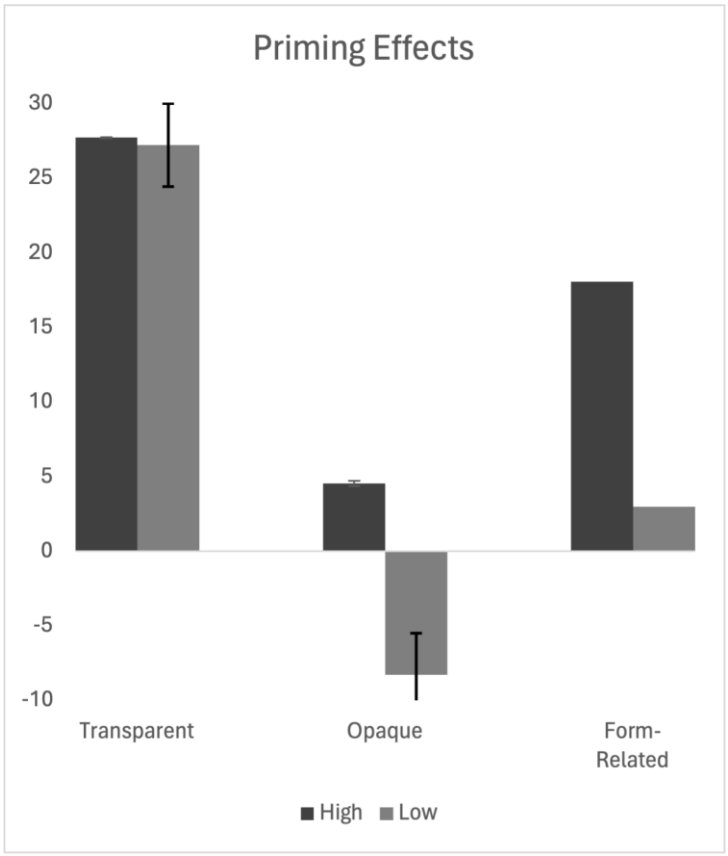


Figure 10. The mean priming effects by condition. Only the transparent effects were significant.

Discussion

Previous research has shown mixed results about the process of morphological priming. This has resulted in conflicting theoretical models that differ on whether the basis of morphological decomposition is purely morphological (morpho-orthographic account as in Rastle et al., 2004), semantic (morpho-semantic account as in Giraudo & Grainger, 2001), or a combination of morphology and semantics (hybrid accounts as in Dipendaele et al., 2009; Morris et al., 2007). Introducing a secondary working memory task – which has been shown to significantly restrict available attentional resources such that true automatic processes are uncovered (Perea et al., 2018) – to morphological priming paradigms may shed some light on these mixed results across the literature.

The present study thus aimed to address the discrepancies in the morphological priming effect literature by introducing a working memory load into the traditional masked priming paradigm. The results yielded two important findings: first, there was a significant main effect of Load across all conditions, which indicates that the working memory task inhibited attentional resources that were available for use under the masked priming conditions. Second, there was a main effect of Relatedness in the transparent condition (misty-MIST), but not the opaque (mister-MIST) or form-related (mistake-MIST) conditions. This pattern of results closely follows the predictions of the morpho-semantic account (see Figure 11 for a comparison), providing strong evidence for a semantic route of morphological decomposition in visual word processing. In the following section, I will discuss the theoretical implications of these findings, as well as some considerations for future research in this field.

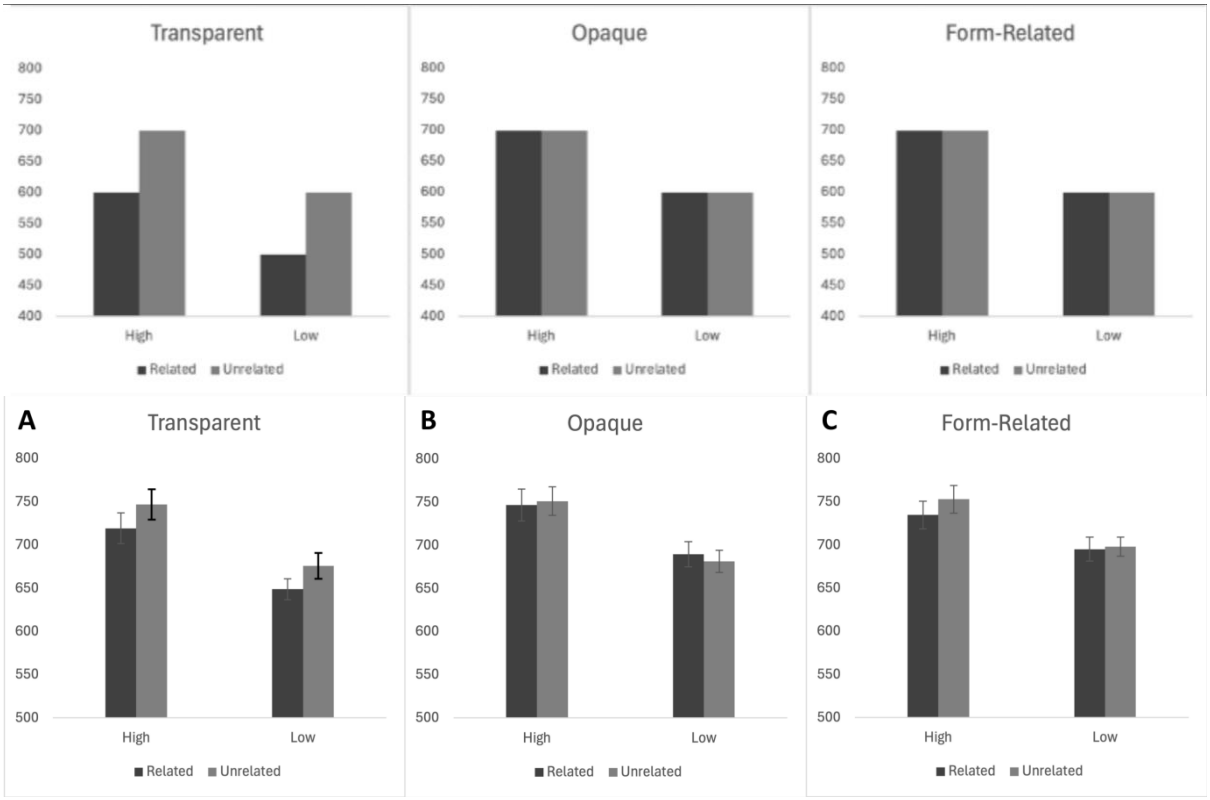


Figure 11. A side-by-side comparison of the predicted results of the present study by the morpho-semantic model (Figure 7B) and the actual results of the present study (Figure 9).

Methodological Differences in Automaticity

There was a significant main effect of load observed across all prime types. As seen in Table 3, there are significantly slower RTs and lower accuracies for the high load condition as compared to the low load condition. This data provides additional support to previous studies which have shown that introducing a secondary working memory task to a masked paradigm can actively inhibit certain additional attentional resources (Perea et al., 2018) and hence yield responses that are entirely automatic in nature. It is clear that the masked priming paradigm does not effectively control for all conscious processing – there are indeed some attentional resources being utilized to process the prime, which subsequently affects responses to targets.

This problem of automaticity in the masked priming paradigm may account for some of the discrepancies seen between the results of the studies which provide support for the

morpho-orthographic account and those which support the morpho-semantic account. It may very well be possible that the assumption of automaticity has led to certain, systematic methodological differences which affected the morphological priming effect. The variety of prime durations (40ms, Rastle et al., 2004; 48ms, Feldman et al., 2004; 50ms, Beyersmann et al., 2012) used across morphological priming studies, combined with the well-established variability of individual differences in masked priming (Boy & Sumner, 2014) and the now-apparent link between factors like working memory, priming and automaticity (Heyman et al., 2015; Perea et al., 2018) pose some serious considerations for the interpretation of previous studies using the traditional masked priming paradigm.

So far, the explanations for the discrepancies between the morpho-orthographic and morpho-semantic accounts have been attributed to methodological differences regarding the selection and control of stimuli. Feldman, O'Connor and Moscoso del Prado Martin (2009) asserted that studies using the stimuli from Rastle et al. (2004) reported morpho-orthographic results due to poorly controlled stimulus properties. On the other hand, Davis and Rastle (2010) attribute the morpho-semantic results observed by Feldman, O'Connor and Moscoso del Prado Martin (2009) to arbitrary orthographic differences between form-related and opaque prime-target pairs. In order to circumvent these issues, the present study derived stimuli from multiple sources, including studies that reported data supporting the both morpho-semantic and morpho-orthographical effects. The stimuli were then further controlled for and matched both within and across conditions over a range of properties (see Table 2). The distinct methodological differences between the present study and extant literature, on the count of stronger control for stimulus properties and automaticity, provides for a more comprehensive set of data regarding the processes of morphological priming. In particular, the data from this study strongly supports the morpho-semantic account, as I will discuss in the next section.

Semantic Basis of Morphological Decomposition

In this study, a significant effect of Relatedness is observed only for the transparent (misty-MIST) prime type; there is no effect of Relatedness observed for the opaque (mister-MIST) or form-related (mistake-MIST) conditions across high or low load conditions. Moreover, the lack of interaction effect between Relatedness and Load in the transparent condition suggests that the priming advantage of transparent primes occurs irrespective of the presence of a memory load, demonstrating that transparent priming is entirely automatic. We can conclusively say that transparent priming occurs even when no conscious attentional resources are diverted to it.

Implications of the Morpho-Semantic Account

The pattern of results closely resembles the predictions of the morpho-semantic model (see Figure 11), providing strong evidence to the claim that the nature of morphological decomposition is automatic in the semantic domain. In other words, the results of this study show that morphological decomposition occurs automatically on the basis of semantics. The finding that morphological priming occurs for transparent primes allows us to shed further light on the cognitive processing of words and, by association, structure of the mental lexicon. The hypotheses of the morpho-semantic model (Giraudo & Grainger, 2001), which are supported by this paper, posit a highly efficient lexical system: in the span of a mere 42ms, the lexical system can register a transparent prime as a whole word (‘misty’), then check whether the internal composition of the word lends itself to semantic transparency (yes), and finally conduct morphological decomposition by stripping the prime down to its stem (‘mist’ + ‘y). Opaque and form-related prime types do not incur any processing advantage due a complete lack of semantic relatedness – they ‘fail’ the initial semantic transparency test, and hence take a longer time to be processed as whole words by the system.

We can extrapolate from this description of the morpho-semantic model that morphological priming in the system is primarily conducted based on semantic relatedness, then decomposed on a morphological basis. The processing advantage occurs because of the recognition of the internal semantic structure of the prime (mist + y), which upon decomposition, allows the participants to activate the target stem (MIST) faster when it is presented. Importantly, this morpho-semantic decomposition process is not only rapid, but also entirely unconscious – this process will occur without any attention or conscious effort exerted by a person.

Considerations for Other Accounts

The results of the present study cannot be supported by the morpho-orthographic view. The complete lack of priming for the opaque condition is opposed to the predictions put forth by Rastle et al. (2004): they suggested that in the early stages of visual processing, prior to conscious processing, an immediate morphological decomposition of a given prime into its constituent morphemes is elicited with no reference to its semantic properties (‘mist’ + ‘y’; ‘mist’ + ‘er’). Priming will not be seen in form-related pairs (mistake-MIST) due to a ack of morphological relatedness. However, if only morphological properties were required for the decomposition process, both the transparent (misty-MIST) and the opaque (mister-MIST) conditions would see a priming effect. But no such effect was observed for the opaque prime type in the present study. I have argued in the previous section that the results of morpho-orthographic study might be explained by the presence of methodological constraints, particularly the use of some latent attentional resources in the masked priming paradigm.

Hybrid accounts may accommodate the results of the present study slightly better with some adjustments. The conventional hybrid accounts suggest that morphological and semantic relatedness each have additive priming effects – that is, form-related pairs (mistake-MIST) will experience no priming effects, but opaque primes (mister-MIST) experience

some priming due to morphological relatedness, and transparent (misty-MIST) pairs experience the greatest priming due to both morphological and semantic relatedness. This graded effect would be seen if decomposition occurred in both morphological and semantic routes parallelly; however, this is not the case for the present study. However, it is possible for hybrid accounts to accommodate these results under a framework that advances a whole word route, in lieu of a morphological route, where both form-related and opaque words are processed, while semantically transparent words are processed through the significantly more efficient semantic route (Beyersmann et al., 2012). A purely morphologically-driven decomposition route, as predicted by the morpho-orthographic account and some hybrid accounts, should show differential priming effects for opaque and form-related conditions, which is not supported by the current data.

Conclusions and Future Directions

The results of the current study yielded strong evidence for the morpho-semantic account, providing some clarity on a controversial debate spanning decades of research in the field of morphology. The pattern of results followed the predictions of the morpho-semantic account even when a secondary memory load task was introduced to the masked priming paradigm, providing compelling support for the claim that morphological decomposition automatically considers semantic properties of a word over purely morphological properties. Additionally, this study makes a case for the presence of potential attentional resources that may affect the unconscious automatic processing which is assumed to be used in a traditional masked priming paradigm. Collectively, this thesis has demonstrated that suspending conscious processing adding a secondary working memory task is methodologically different from the masked priming paradigm; further, this suspending of conscious processing generates a strong effect of semantic transparency in morphological priming.

Subsequent research must hence carefully consider whether the data obtained using the traditional masked priming paradigm reflects purely automatic processes. It may be fruitful to consider the comparison of a priming effect under a high load, low load and no load condition to uncover the precise conscious processing mechanisms that seep into the masked priming paradigm. The findings from such a study would be able to better inform the gap between the morpho-orthographic and morpho-semantic accounts. It may also be useful to examine whether there the effects of individual differences such as vocabulary, spelling and age of acquisition observed in other studies (Andrews & Lo, 2013; Milin et al., 2017) also contributes to the semantic transparency effect under a dual task paradigm similar to the one presented in this study. Such findings could shed more insight into the mechanisms of the models of morphological priming and the broader discipline of lexical processing.

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Appendices

Appendix A: List of Stimuli Used in the Experiment

Form-Related			Opaque			Transparent		
Prime		Target	Prime		Target	Prime		Target
Unrelated	Related		Unrelated	Related		Unrelated	Related	
gazelle	rugby	RUG	cracker	fairy	FAIR	slowly	heroic	HERO
manicure	costume	COST	penance	limber	LIMB	drunkard	slowly	SLOW
pillow	glade	GLAD	hostage	plumage	PLUM	heroic	cynical	CYNIC
window	decadent	DECADE	drawer	early	EARL	fussy	package	PACK
catch	pasta	PAST	helmet	treaty	TREAT	cynical	risky	RISK
plush	pillow	PILL	early	metal	MET	greenery	wrecker	WRECK
scandal	gazelle	GAZE	plumage	inventory	INVENT	cheerful	editor	EDIT
heaven	manicure	MANIC	fairy	facetious	FACET	cloudy	fussy	FUSS
ethereal	dialect	DIAL	facetious	cracker	CRACK	editor	drunkard	DRUNK
glade	plush	PLUS	trumpet	hostage	HOST	mixer	hanger	HANG
costume	window	WIND	treaty	helmet	HELM	herbal	golfer	GOLF
decadent	electron	ELECT	factory	mission	MISS	golfer	lonely	LONE
humble	catch	CAT	metal	flourish	FLOUR	acreage	creepy	CREEP
regiment	humble	HUM	inventory	penance	PEN	painless	cloudy	CLOUD
cardinal	stunt	STUN	choppy	brandy	BRAND	risky	herbal	HERB
rugby	ethereal	ETHER	flourish	trumpet	TRUMP	shameful	cheerful	CHEER
dialect	heaven	HEAVE	limber	question	QUEST	eating	mixer	MIX
wallow	freeze	FREE	Fatal	drawer	DRAW	lonely	shameful	SHAME
stunt	scandal	SCAN	poster	factory	FACT	hanger	creamy	CREAM
freeze	wallow	WALL	brisket	hinder	HIND	package	greenery	GREEN
hotel	regiment	REGIME	question	beaker	BEAK	creepy	eating	EAT
electron	hotel	HOT	mission	Fatal	FAT	fixing	painless	PAIN
restrict	fluid	FLU	hinder	poster	POST	creamy	robber	ROB
pasta	cardinal	CARD	brandy	brisket	BRISK	robber	fixing	FIX
fluid	restrict	REST	beaker	choppy	CHOP	wrecker	acreage	ACRE
monkey	extract	EXTRA	summer	secretary	SECRET	employer	eruption	ERUPT
quartz	monkey	MONK	tower	easter	EAST	faithful	employer	EMPLOY
squawk	pulpit	PULP	secretary	cranky	CRANK	healer	guilty	GUILT
pulpit	scrape	SCRAP	rampage	former	FORM	sewed	harder	HARD
grimace	napkin	NAP	putty	shower	SHOW	moisture	insistent	INSIST
plaintiff	arctic	ARC	literal	jetty	JET	foolish	moisture	MOIST
sternum	tooth	TOO	cranky	archer	ARCH	adopted	raider	RAID
arctic	spinach	SPIN	supplement	porter	PORT	dimly	leakage	LEAK
button	plaintiff	PLAIN	bullet	massage	MASS	raider	boiler	BOIL
brothel	studio	STUD	customer	ample	AMP	boiler	holder	HOLD

termite	squawk	SQUAW	massage	naughty	NAUGHT	boldness	swimmer	SWIM
single	brothel	BROTH	easter	literal	LITER	lender	healer	HEAL
studio	quartz	QUART	master	putty	PUTT	insistent	foolish	FOOL
spinach	sparse	SPAR	rational	pigment	PIG	marker	faithful	FAITH
surgeon	single	SING	jetty	infantry	INFANT	sailor	leader	LEAD
easel	grimace	GRIM	naughty	tower	TOW	leakage	marker	MARK
fortune	galaxy	GALA	former	rational	RATION	leader	aimed	AIM
tooth	fortune	FORT	shower	supplement	SUPPLE	guilty	adopted	ADOPT
extract	termite	TERM	archer	rampage	RAMP	harder	loafer	LOAF
tartan	button	BUTT	porter	fasten	FAST	loafer	sewed	SEW
galaxy	tartan	TART	buzzard	master	MAST	swimmer	sailor	SAIL
sparse	surface	SURF	pigment	bullet	BULL	renewable	lender	LEND
napkin	easel	EASE	ample	customer	CUSTOM	eruption	dimly	DIM
surface	sternum	STERN	fasten	summer	SUM	holder	boldness	BOLD
scrape	surgeon	SURGE	infantry	buzzard	BUZZ	aimed	renewable	RENEW
bandit	begin	BEG	plucky	punish	PUN	binder	poetry	POET
textile	twinkle	TWIN	grateful	coaster	COAST	deeply	player	PLAY
ginger	sight	SIGH	fallible	wander	WAND	seller	flexible	FLEX
pumpkin	think	THIN	crooked	bunker	BUNK	painter	bulbous	BULB
twinkle	strident	STRIDE	hearty	discern	DISC	dietary	farmer	FARM
wanton	against	AGAIN	every	sneaker	SNEAK	poacher	joyful	JOY
villain	wanton	WANT	punish	witness	WIT	joyful	medical	MEDIC
candidacy	dollop	DOLL	bunker	crooked	CROOK	blender	viewer	VIEW
strident	pumpkin	PUMP	sandal	bracelet	BRACE	flexible	seller	SELL
grammar	curtain	CURT	sneaker	audition	AUDIT	rider	blender	BLEND
tease	ginger	GIN	bracelet	flicker	FLICK	drainage	fluffy	FLUFF
salmonella	crown	CROW	wander	grateful	GRATE	rejection	cruelly	CRUEL
phonetic	china	CHIN	trolley	vanish	VAN	poetry	bomber	BOMB
think	phonetic	PHONE	courteous	stringent	STRING	fluffy	rejection	REJECT
yellow	salmonella	SALMON	flicker	ponder	POND	opened	verbal	VERB
bushel	textile	TEXT	corner	planet	PLAN	cruelly	opened	OPEN
crown	inferno	INFER	vanish	plucky	PLUCK	medical	dietary	DIET
curtain	grammar	GRAM	discern	every	EVER	stalker	welder	WELD
skirt	yellow	YELL	crabby	trolley	TROLL	farmer	deeply	DEEP
begin	skirt	SKI	stringent	crabby	CRAB	welder	poacher	POACH
sight	villain	VILLA	planet	courteous	COURT	player	rider	RIDE
dollop	candidacy	CANDID	witness	fallible	FALL	bomber	painter	PAINT
china	bushel	BUSH	audition	hearty	HEART	viewer	drainage	DRAIN
against	bandit	BAND	coaster	corner	CORN	bulbous	binder	BIND
inferno	tease	TEA	ponder	sandal	SAND	verbal	stalker	STALK
lesson	shunt	SHUN	message	flower	FLOW	walked	badly	BAD
smuggle	general	GENE	filly	saucer	SAUCE	hunter	chilly	CHILL
general	wonder	WON	flower	tailor	TAIL	filthy	gloomy	GLOOM

tactile	lesson	LESS	united	dormant	DORM	locker	mainly	MAIN
stirrup	nickel	NICK	charter	luster	LUST	zealous	hunter	HUNT
whimper	harmony	HARM	tender	rodent	ROD	bearded	flashy	FLASH
turnip	shovel	SHOVE	irony	thicket	THICK	looter	bearded	BEARD
earth	tipsy	TIP	listen	liquidate	LIQUID	badly	buying	BUY
harmony	whimper	WHIM	crater	fleeting	FLEET	flashy	bearable	BEAR
stampede	palace	PAL	liquidate	rustic	RUST	pitcher	faulty	FAULT
dragon	shallow	SHALL	dormant	scary	SCAR	floater	looter	LOOT
palace	colonel	COLON	saucer	solvent	SOLVE	runner	floater	FLOAT
shovel	tinsel	TIN	rustic	bowler	BOWL	engineer	pitcher	PITCH
colonel	stampede	STAMP	scary	charter	CHART	scalding	shipment	SHIP
tipsy	turnip	TURN	rodent	organic	ORGAN	buying	engineer	ENGINE
wonder	tactile	TACT	tailor	listen	LIST	critical	marshy	MARSH
nickel	stirrup	STIR	luster	united	UNIT	mainly	dirty	DIRT
twitch	earth	EAR	bowler	sniper	SNIP	marshy	walked	WALK
bunch	command	COMMA	fleeting	irony	IRON	chilly	scalding	SCALD
bellow	forceps	FORCE	thicket	message	MESS	reader	zealous	ZEAL
forceps	smuggle	SMUG	clover	spanner	SPAN	bearable	filthy	FILTH
command	bellow	BELL	sniper	tender	TEND	shipment	locker	LOCK
tinsel	twitch	TWIT	solvent	crater	CRATE	gloomy	critical	CRITIC
shallow	bunch	BUN	spanner	filly	FILL	dirty	runner	RUN
shunt	dragon	DRAG	organic	clover	CLOVE	faulty	reader	READ

Nonwords

Prime			Prime			Prime		
Unrelated	Related	Target	Unrelated	Related	Target	Unrelated	Related	Target
blushed	eloped	ELIPE	hiding	embedded	INBETE	cheated	ugly	AGLA
gleeful	mailed	GAIL	crispy	pawed	FAW	headed	groomed	GROON
potted	slammed	SNAN	helmed	swimming	SWAW	crusty	cheated	CHEAR
hellish	blushed	BLUS	forgeting	clearly	CLOUR	boomed	nabbed	RAB
stmmed	ashy	ESHES	shelved	spotted	SPOP	cultish	lapped	EAP
flying	napped	WAPS	frosty	fitted	MIT	poked	speared	SPEAM
justly	bitten	WITE	soaked	baited	FAIT	lapped	lagged	PAGS
gritty	lofty	FOFT	dotted	ribbed	PIB	dazed	purred	SER
cropped	gritty	GRIR	clotted	tuned	BUNE	explained	incensed	INCUST
salted	spitting	SPAG	mostly	combed	FOMB	dearest	balding	HALD
starkly	faded	TADE	ribbed	dotted	DOW	eluded	fouled	HOUL
merging	edging	ELSTA	draped	hater	HOTER	dozed	worked	HORK
cribbing	lately	WAME	kidding	broken	BRACKER	biked	feasted	SEST
noted	upgraded	STRADE	pounded	trotted	POLT	greying	piper	BIPE
charmed	unworthy	UNVER	broken	tanked	FANK	duelled	selfish	VELF
lately	riddled	GID	conned	audited	AUNOT	doubled	scabbed	SCAY
agility	primly	PRIN	uttered	escaped	BRAPE	flocked	dreadful	DREND
banned	penalised	MEA	gated	reached	HEACH	rejoined	tanned	OTAN

stressed	costly	CORTA	trotted	winning	WUN	toasted	wrathful	DRATH
flailed	ruined	RURN	losing	shipped	THIP	holding	moving	MOVIC
safely	chucked	NUCK	braked	fishy	LISH	dsealed	biked	RIKE
prayed	blissful	BLIM	swinging	treated	TREET	layered	womanly	WAMB
husky	carved	FOLFS	darkly	shocked	THOCK	croaked	peaceful	BEACE
carved	sunny	HON	crammed	ceded	MEDE	loudly	rinsed	FINSE
embedded	soaked	BOAK	speared	waxed	WIX	peaceful	meaty	LEAT
crumbly	braked	CLATE	milky	boomed	GOMB	ashy	running	RANIF
limply	helmed	EALM	waxed	chimed	CHAME	rotted	polled	COLL
sloppy	banking	MANK	hitting	taped	PAPE	meaning	stmmed	STUM
fuzzy	subtler	SUDFLE	taped	cultish	FULT	napped	justly	PUST
grandly	rarely	RAREEN	slotted	poked	LOKE	mailed	rotted	ROG
offered	saintly	NAINTS	groomed	caned	HANE	eloped	flying	FLI
making	grandly	CRAND	missed	repayed	REPEY	piled	hellish	PELL
hater	clotted	CLOW	reeked	locked	AWK	kissed	regarded	REGARN
banking	mostly	OOST	juicy	reeked	REAK	faded	starkly	STERK
subtler	inhaled	INHATE	purred	maimed	FAIMS	shaped	foggy	FOO
patted	sloppy	BURB	balked	gulped	BULP	talked	salted	EALT
scathing	conned	VON	selfish	valued	PALVE	guarded	flipped	FLIN
creaky	trusted	TROST	wolfish	greying	GREX	madness	talked	TELK
mossy	flaked	FLATE	valued	soupy	DOUP	curbed	richly	SICH
reached	roosted	COOST	gulped	balked	FALK	penalised	noted	GNONE
audited	flagged	FLAB	stalled	briefly	VIEF	fried	famed	BAME
shrinking	blackness	BLUCK	foxy	holding	DELD	unworthy	jealousy	JELLA
tanked	pounded	COUND	muted	loudly	ROUD	jealousy	cured	CUPE
droves	plowed	FLOY	acidic	hurting	DURT	knotted	betting	BEP
roared	chalky	CHARK	furry	acidic	FLIDIC	gravely	coping	COBIN
claimed	slimness	SLAN	reeled	scarcely	SCARD	rapped	messy	BESH
slimness	losing	ROST	womanly	leafy	BEAF	blissful	merely	GERE
barked	lifted	LIFEED	aided	chided	CHIRE	chucked	stopped	STOR
chalky	used	USAT	slapped	slapped	SHAP	stopped	sluggish	SKUG
shipped	roared	ROR	rammed	aided	PIDES	peeled	toyed	TIY
pointed	inanely	INAN	regarded	kissed	HUS	finish	burned	BUR
belted	swinging	FING	bitten	zesty	MEST	toyed	finish	FIP
shocked	barked	TARK	zesty	sliding	GLID	poshly	breaded	BREAT
ceded	loved	BOBE	slammed	degraded	DEGACE	pawed	frosty	FREST
chimed	knocked	KNECK	paced	potted	MOT	fateful	slacked	SPACK
caned	milky	RILK	polled	paced	CACE	clasped	silky	LILK
knocked	lyrical	BYRIC	planted	cropped	CRUD	clearly	crispy	CRISK
lagged	greedy	GREEP	pulpy	madness	NAD	gruffly	crumbly	PRUM
ugly	slotted	SLOY	flipped	pulpy	BULF	inhaled	patted	JAT
halved	crusty	DRUST	spitting	merging	TURGE	buttoned	fashioned	FASH
nabbed	headed	HEDA	shrewdly	brassy	BROSS	combed	frothing	POTH

exited	shelved	HELF	upgraded	fried	FRI	fashioned	fuzzy	PUZZ
flooded	hitting	HIX	tested	naked	PUDE	saintly	draped	MAPE
cooling	lamely	LEMP	honked	charmed	CHAM	rarely	gambled	AMBS
lamely	cooling	LOAL	surely	speaking	SPOW	flagged	creaky	PREAK
locked	passed	MORCE	cured	tested	TEWT	cutely	kidding	RIT
adeptly	dozed	BOZE	primly	honked	DONK	winning	cutely	BUTE
tanned	doubled	CHUET	discussed	knotted	KNIN	flaked	gated	WATE
scabbed	stalled	STAWL	mulish	surely	SUREON	trusted	scathing	SATH
chided	croaked	CROAT	coping	stressed	STASS	inanely	fateful	BRATE
rinsed	vaguely	MAUGE	sunny	flailed	FLALL	dunked	darkly	DARM
insanely	layered	FAYER	plotted	peeled	EAL	used	belted	TELT
swooned	decked	LECK	leaving	rapped	RAL	lyrical	exited	EXIL
gladly	dawned	MAWN	ruined	plotted	PLIT	greedy	flooded	FLOUD
decked	muted	HUTE	burned	prayed	PRAG	trimmed	explained	EXPLOM
degraded	gleeful	BLEE	shopped	gravely	BROVE	repayed	eluded	ELUBE
running	piled	PIYL	breaded	limply	HIMP	veiled	juicy	GUICE
petted	meaning	HEANT	padded	halved	HERF	maimed	dazed	SAZE
sliding	petted	CET	swimming	padded	PED	fouled	downy	BOWN
trapped	lucky	PUCT	baited	poshly	HOSH	downy	missed	DIS
gasped	shaped	APH	fitted	gruffly	TRUFF	calmed	veiled	FEIL
foggy	chunky	MUNK	pinkish	forgeting	FOGET	worked	barred	MARN
chunky	shrewdly	SHRAW	silky	hiding	MIDE	passed	adeptly	ADAPE
richly	planted	PLARK	regretful	digging	SUG	balding	trimmed	FRIM
brassy	gasped	GASO	gambled	regretful	MEGRET	barred	calmed	RALM
lofty	guarded	QUARD	frothing	offered	AFF	incensed	dearest	LEAR
lucky	trapped	THRAP	tuned	making	RADE	feasted	toasted	POAST
edging	curbed	SURB	digging	buttoned	BOSTOM	wrathful	dsealed	SIEL
famed	mulish	BULE	roosted	cried	CRIEK	dreadful	flocked	FLULE
bathing	discussed	SISK	escaped	uttered	ULER	soupy	duelled	DOEL
speaking	cribbing	CRIN	cried	shrinking	STRIMP	briefly	rejoined	REJORN
betting	banned	BANNID	blackness	kindled	DILN	moving	wolfish	BOLF
naked	bathing	YTHE	willing	mossy	COSS	piper	foxy	JOX
riddled	agility	AGIVE	kindled	willing	WHAN	meaty	furry	RUR
merely	leaving	LEFF	treated	pointed	SOINT	fuelled	swooned	SWOOM
messy	safely	TAFE	lifted	crammed	PRAR	vaguely	rammed	BAM
sluggish	husky	HASK	urently	droves	CROVE	scarcely	insanely	ENSONE
costly	shopped	SHON	loved	claimed	SAIMS	leafy	gladly	FLAND
slacked	clasped	FASP	fishy	dunked	LUNK	dawned	reeled	ZEEL
spotted	pinkish	PONK	plowed	urently	URGON	hurting	fuelled	BUE

Appendix B: Email of Approval for DERC Application

RE: [Submission] DERC Application - 2023-November-10 - Malubhoy Alefiya

Psych Ethics Enquiries <psybox27@nus.edu.sg>
Thu 11-Jan-24 16:58
To:Malubhoy Alefiya <alefiya.m@u.nus.edu>
Cc:Melvin J. Yap <melvin@nus.edu.sg>

2 attachments (704 KB)
DERC-edited.pdf; DERC Reporting Template V5 2023-05-19.xls;

Dear Alefiya

This is to inform you that your application for DERC review entitled “Effects of Working Memory on Morphological Priming” has been approved by the DERC.

Please note:
- Psych-DERC Reference Code: 2023-November-10
- Please use this e-mail as approval for your request.